Paper Code:

Measurement-Oriented Microscopy Reconstruction for Cross-Magnification Micro-Displacement Inspection

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Introduction

Optical inspection estimates motion from image features. A sharper reconstructed edge is useful when its measured position remains stable. Optical blur, pixel sampling and reconstruction-induced boundary shifts can all change the reported displacement, especially when an inspection workflow switches objectives.

MOSR combines U-Net context [1], calibrated optical guidance and anti-aliased processing [2] to address these effects. Bounded correction retains the original image grid and limits geometry changes while improving boundary visibility. One fixed set of learned weights serves both objectives.

At a common nominal step of 1 µm, this study compares Raw, fixed Wiener and MOSR at 20× and 50×. Each condition contributes 30 measured single steps. The central outcome is the variation of these steps, linking reconstruction quality directly to displacement measurement.

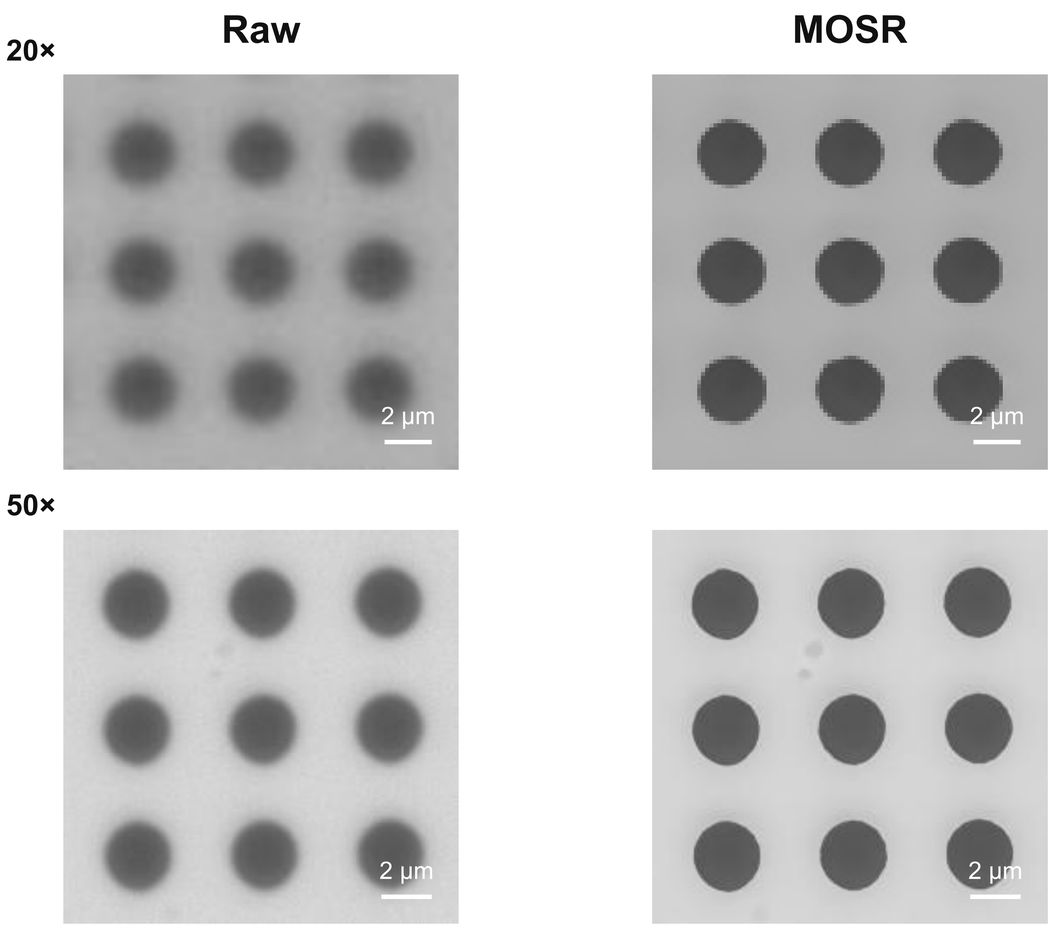


Fig. 1. Raw and MOSR images from the measured 20× and 50×, 1 µm datasets. Paired images use identical fields and grayscale ranges. Scale bars: 2 µm.

Methods

Data and reconstruction. At 20×, 31 frames form 30 consecutive steps; 50× uses the original 30 locked adjacent pairs. Both conditions employ a 1 µm nominal step. MOSR uses fixed weights and native-grid outputs, with 25 measured PSF kernels at 20× and one calibrated kernel at 50×. Fixed Wiener uses a regularization value of 0.0565.

Boundary and center. A coarse active contour [3] initializes each pillar boundary. Intensity profiles normal to the contour use 0.1-pixel spacing. Quadratic refinement locates each gradient maximum between samples. Geometric least-squares circle fitting [4] uses 64 refined points at 20× and 96 at 50× to estimate the center.

Displacement and statistics. Coordinate-wise medians of nine pillar-center changes define the array shift. Its component along the acquisition motion direction gives the measured step. Object-space conversion uses the scale in Table I. Each condition yields one sample SD from its 30 steps. Raw, Wiener and MOSR share the same images, features and center-calculation settings.

TABLE I. SEPARATE ACQUISITION CONDITIONS

Objective	Step (µm)	Measured steps	µm/pixel
20×	1.0	30	0.1725
50×	1.0	30	0.0690

Step SD uses all 30 displacements with denominator 29. Reduction (%) = 100 × (1 − SD(MOSR) / SD(Raw)). Each trace plots measured step minus the nominal 1 µm; each objective is summarized separately.

Results

At a common 1 µm step, MOSR lowers displacement variation at both objectives. Results are calculated within each acquisition condition.

20×, 1 µm: 30 single steps

57.2%

Reduction in Step SD

9.506 nm Raw
4.068 nm MOSR

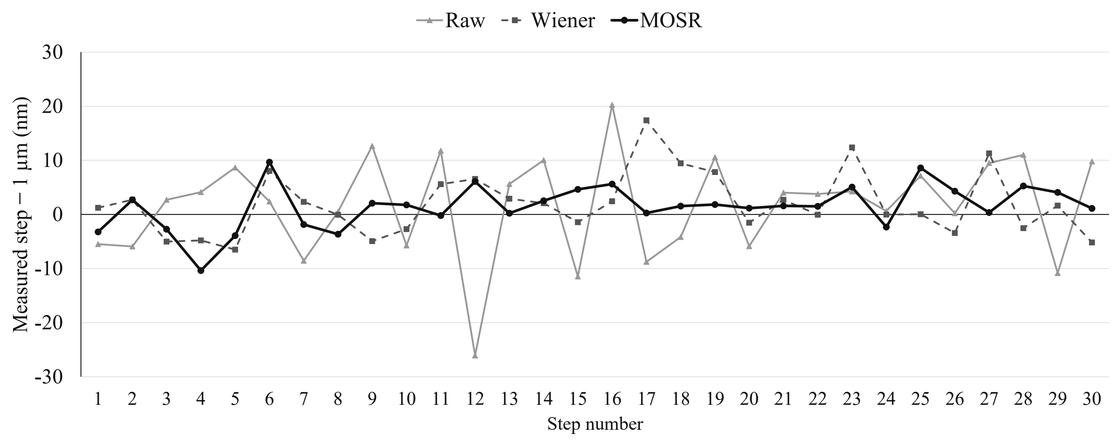


Fig. 2. Thirty single-step displacements at 20×. The traces show measured step minus 1 µm, using the same axes and units as the 50× experiment.

At 20×, Step SD decreases from 9.506 nm for Raw to 5.794 nm for fixed Wiener and 4.068 nm for MOSR. MOSR reduces the variation by 57.2% relative to Raw.

50×, 1 µm: 30 single steps

23.5%

Reduction in Step SD

5.938 nm Raw
4.542 nm MOSR

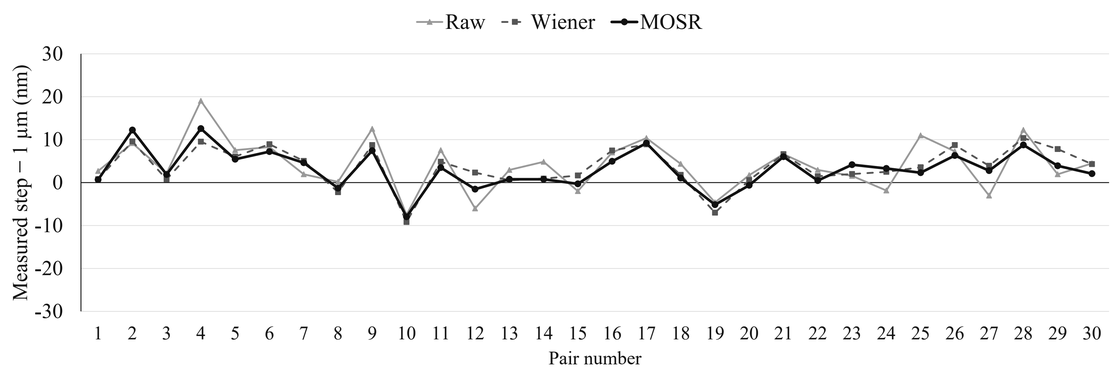


Fig. 3. Thirty locked single-step displacements at 50×. Step SD is 5.938 nm for Raw, 4.722 nm for fixed Wiener and 4.542 nm for MOSR.

Conclusions

MOSR reduces the SD of 30 measured single steps by 57.2% at 20× and 23.5% at 50×, at a common nominal step of 1 µm. Each result retains its own acquisition condition.

One reconstruction model supports both objectives with calibrated optical inputs. Clearer boundaries and lower step variation support cross-magnification optical inspection.

Bibliography

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